1

2

3

4

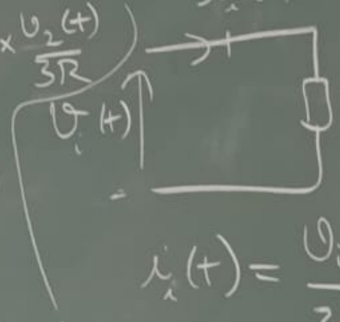
5

$$p_2(t) = \frac{2V^2}{R}\sin^2(\omega t) = f(t)$$

En 3φ

$$p_2(t) = v_1(t)\,i_1(t) + v_2(t)\,i_2(t) + v_3(t)\,i_3(t)$$

$$p_2(t) = v_1(t)\times\frac{v_1(t)}{3R} + v_2(t)\times\frac{v_2(t)}{3R} + v_3(t)\times\frac{v_3(t)}{3R}$$



$$i_1(t) = \frac{v_1(t)}{3R}$$

$$p_2(t) = \frac{1}{3R}\left(v_1^2(t) + v_2^2(t) + v_3^2(t)\right)$$

avec
$$\begin{cases} v_1(t) = V\sqrt{2}\sin(\omega t) \\ v_2(t) = V\sqrt{2}\sin\left(\omega t - \frac{2\pi}{3}\right) \\ v_3(t) = V\sqrt{2}\sin\left(\omega t - \frac{4\pi}{3}\right) \end{cases}$$

6

$$p_2(t) = \frac{2V^2}{3R}\left(\sin^2\omega t + \sin^2\left(\omega t - \frac{2\pi}{3}\right) + \sin^2\left(\omega t - \frac{4\pi}{3}\right)\right)$$

or $\sin^2\theta = \frac{1-\cos 2\theta}{2}$

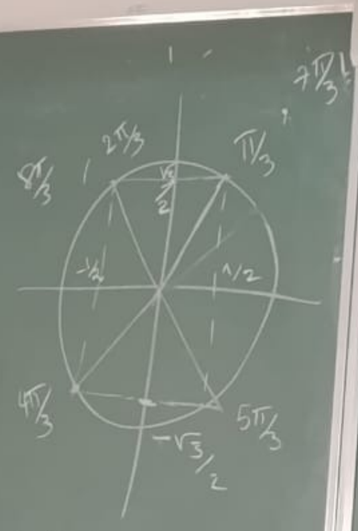
$$p_2(t) = \frac{2V^2}{2\times 3R}\left(1-\cos(2\omega t) + 1-\cos\left(2\omega t - \frac{4\pi}{3}\right) + 1-\cos\left(2\omega t - \frac{8\pi}{3}\right)\right)$$

$$p_2(t) = \frac{V^2}{3R}\left(3 - \cos(2\omega t) - \cos\left(2\omega t - \frac{4\pi}{3}\right) - \cos\left(2\omega t - \frac{8\pi}{3}\right)\right)$$

or $\cos a\cos b + \sin a\sin b = \cos(a-b)$

$$p_2(t) = \frac{V^2}{3R}\left(3 - \left(\cos(2\omega t) + \cos(2\omega t)\times\cos\left(\frac{4\pi}{3}\right) + \cos(2\omega t)\cos\left(\frac{8\pi}{3}\right)\right.\right.$$
$$\left.\left. + \sin 2\omega t\times\sin\left(\frac{4\pi}{3}\right) + \sin(2\omega t)\sin\left(\frac{8\pi}{3}\right)\right)\right)$$

7



$$p_2(t) = \frac{V^2}{3R}\left(3 - \cos(2\omega t)\left(1+\cos\left(\frac{4\pi}{3}\right)+\cos\left(\frac{8\pi}{3}\right)\right)\right.$$
$$\left. - \sin(2\omega t)\left(\sin\left(\frac{4\pi}{3}\right)+\sin\left(\frac{8\pi}{3}\right)\right)\right)$$

$$p_2(t) = \frac{V^2}{3R}\times 3 = \frac{V^2}{R} = cte = P_{moy}$$

8

8) Conclusion

En 3φ équilibré, la puissance instantanée délivrée est constante et égale à sa puissance moyenne, ce qui est idéal pour l'alimentation de charges industrielles, notamment les moteurs électriques ($C_{couple} = cte$). Le circuit monophasé a une puissance instantanée pulsée ce qui peut générer des vibrations (bruits)

et un
régulier.

9

instantanée

sa puissance

l'alimentation

ut les

circuit

née pulsée

bornes)

et un fonctionnement moins régulier.

10

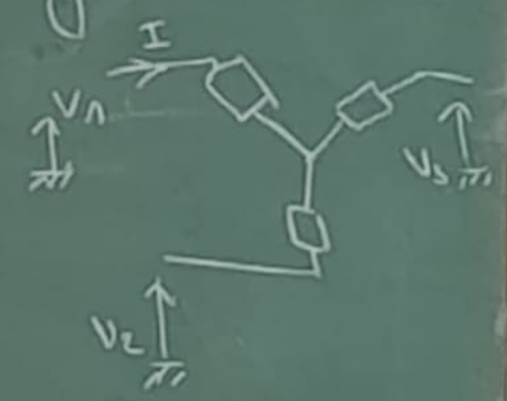
Série 2 (Suite)

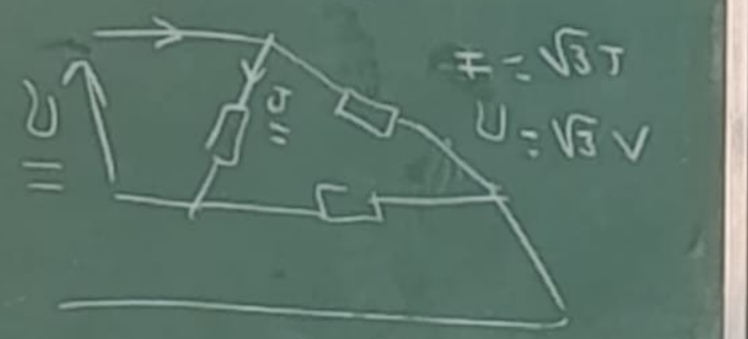
Ex 2

récepteur 3φ R = 100 Ω

230 V / 400 V f = 50 Hz

V U





11

1) I = ? , P = ? en Y

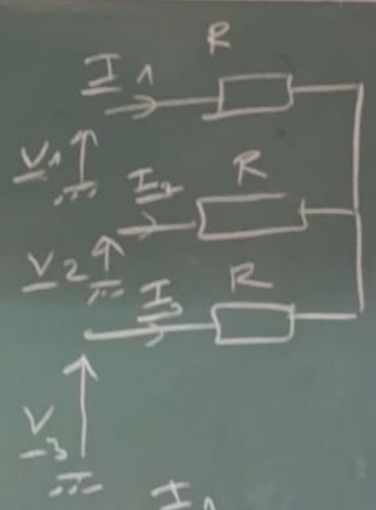
le syst. 3φ est équilibré

$\underline{V}_1 + \underline{V}_2 + \underline{V}_3 = 0$; $\underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 0$

$V_1 = V_2 = V_3$

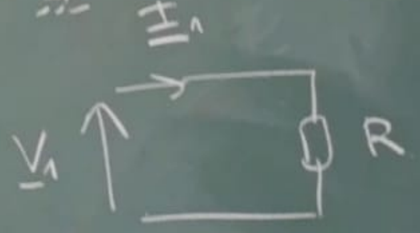
$I_1 = I_2 = I_3$

$\underline{Z}_i = R$



La courant en ligne :

$$\underline{I}_1 = \frac{\underline{V}_1}{R}$$



$$I = I_3 = I_2 = I_1 = \frac{V_1}{R} = \frac{230}{100}$$

$$I = 2{,}3 \text{ A}$$

$$P = \frac{P}{Y} = 3RI^2 = 3\frac{V^2}{R} = 3 \times \frac{230^2}{100} = 1587 \simeq 1{,}6 \text{ KW}$$

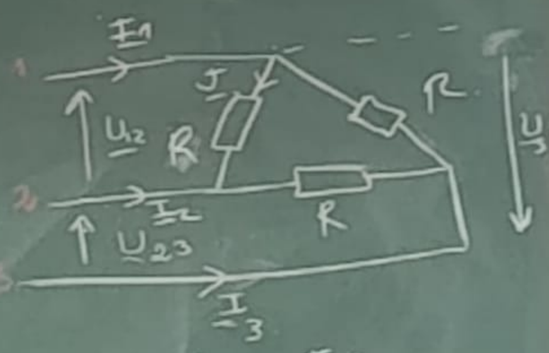
2)

U_{12}

I_1

J_i

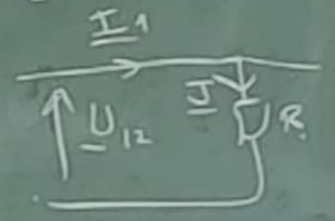
2) $I = ?$, $P = ?$ en Δ



$U_{12} = U_{23} = U_{31} = U$

$I_1 = I_2 = I_3 = I$

$J_i = J$



$$I = \frac{U_{12}}{R}$$

$$J = \frac{U}{R} = \frac{400}{100} = 4\,A$$

$$\Rightarrow \underline{I = \sqrt{3}\,J = 6,93\,A}$$

$$\underline{P = P_\Delta = 3RJ^2 = 3\frac{U^2}{R} = 4,8\,kW}$$

12

3) Conclusion

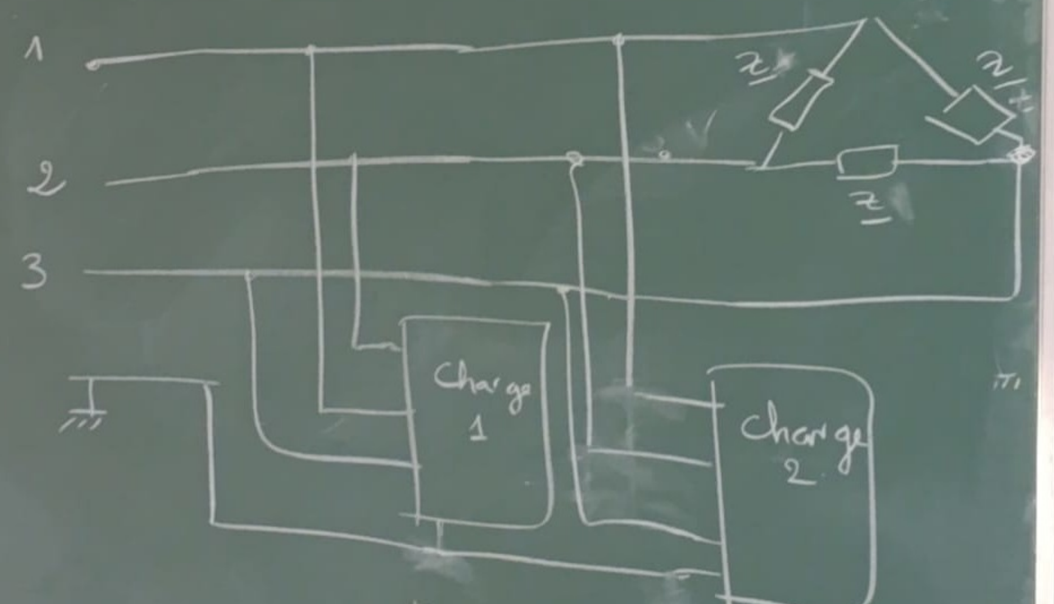
En couplage Δ, le courant est 3 fois supérieur que le courant en ligne en couplage Y, il en est de même pour la puissance active :

$$I_\Delta \simeq 3 I_Y$$

$$P_\Delta \simeq 3 P_Y$$

13

EX3 3φ 230V / 400V



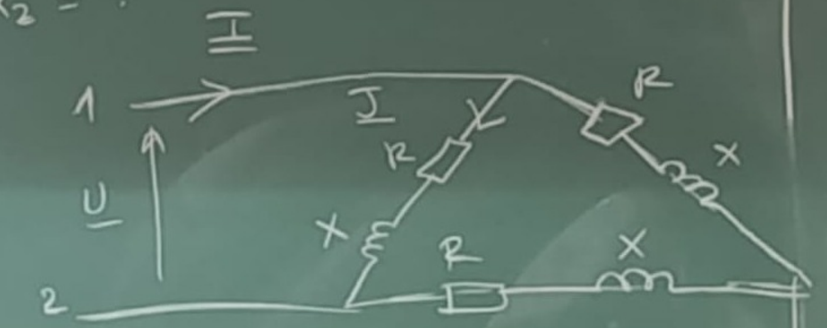
$P_1 = 6\,kW$ $\quad \cos\varphi_1 = 1$

$P_2 = 3 \times 5\,kW$ $\quad \cos\varphi_2 = 0,8$

14

15

1) $P_z = ?$ $\quad Q_z = ?$



$$J = \frac{U}{\sqrt{R^2 + X^2}}$$

$$\underline{Z = R + jX}$$

Z : R (Ra), X (Q)

AN

$$J = \frac{400}{\sqrt{10^2 + 15^2}} = 22,19\,A$$

Donc

$$\underline{P_z = 3RJ^2 = 3 \times 10 \times (22,19)^2 = 14,77\,kW}$$

$$\underline{Q_z = 3XJ^2 = 3 \times 15 \times (22,19)^2 = 22,1\,KVAR}$$

16

2) $P_t = ?$

$P_t = P_1 + P_2 + P_3$

$= 6k + 15k + 14,77k$

$P_t = 35,77\ kw$

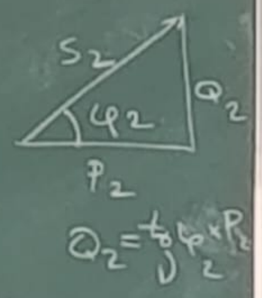
3) $Q_t = ?$

$Q_t = Q_1 + Q_2 + Q_3$

$= P_2 \operatorname{tg}\varphi_2 + Q_3$

$= 0,75 \times 15k + 22,15k$

$Q_t = 33,4\ KVAR$



$Q_2 = \operatorname{tg}\varphi_2 \times P_2$

$\cos\varphi_2 = 0,8$

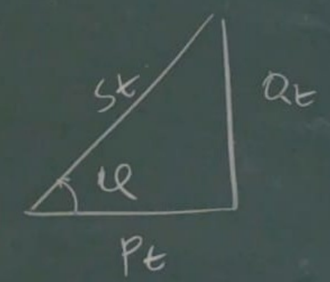
$\varphi_2 = 36,87°$

$\operatorname{tg}\varphi_2 = 0,75$

4) $S_t = ?$, $I = ?$

$S_t = \sqrt{P_t^2 + Q_t^2}$

$S_t = 48,94\ KVA$



$S_t = \sqrt{3}\,U\,I = 3V\,I$

$I = \dfrac{S_t}{\sqrt{3}\,U} = \dfrac{48,94\ 10^3}{\sqrt{3} \times 400}$

$I = 70,64\ A$

17

18

5) $\cos\varphi = ?$

$\cos\varphi = P_t / S_t$

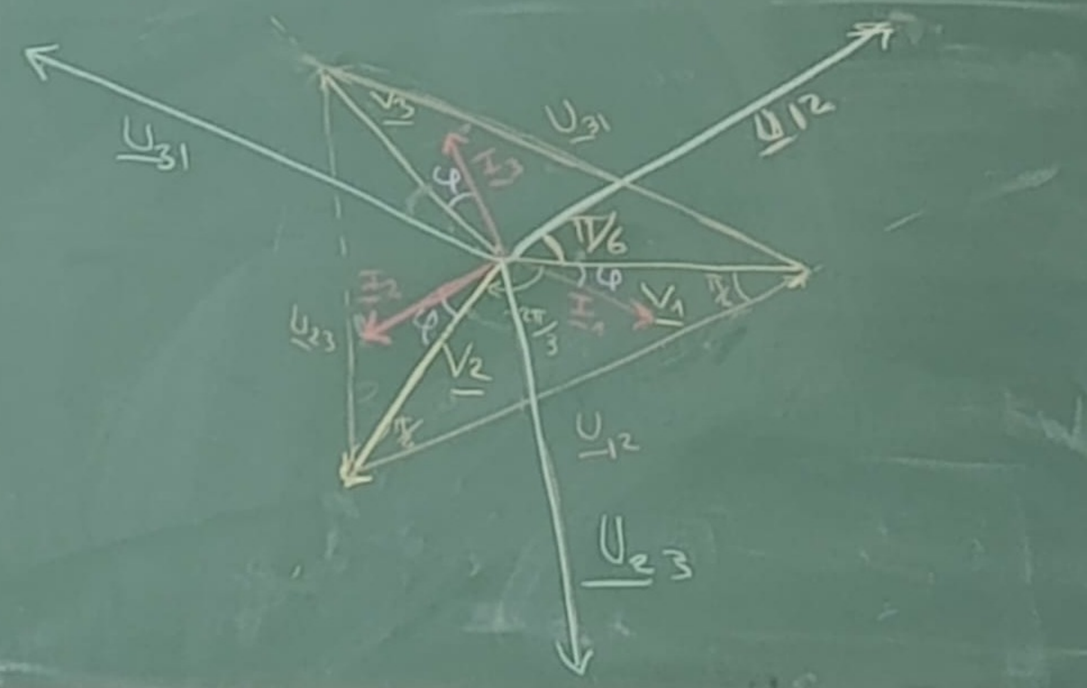
$\cos\varphi = \dfrac{35,77}{48,94} = 0,731 < 0,8$

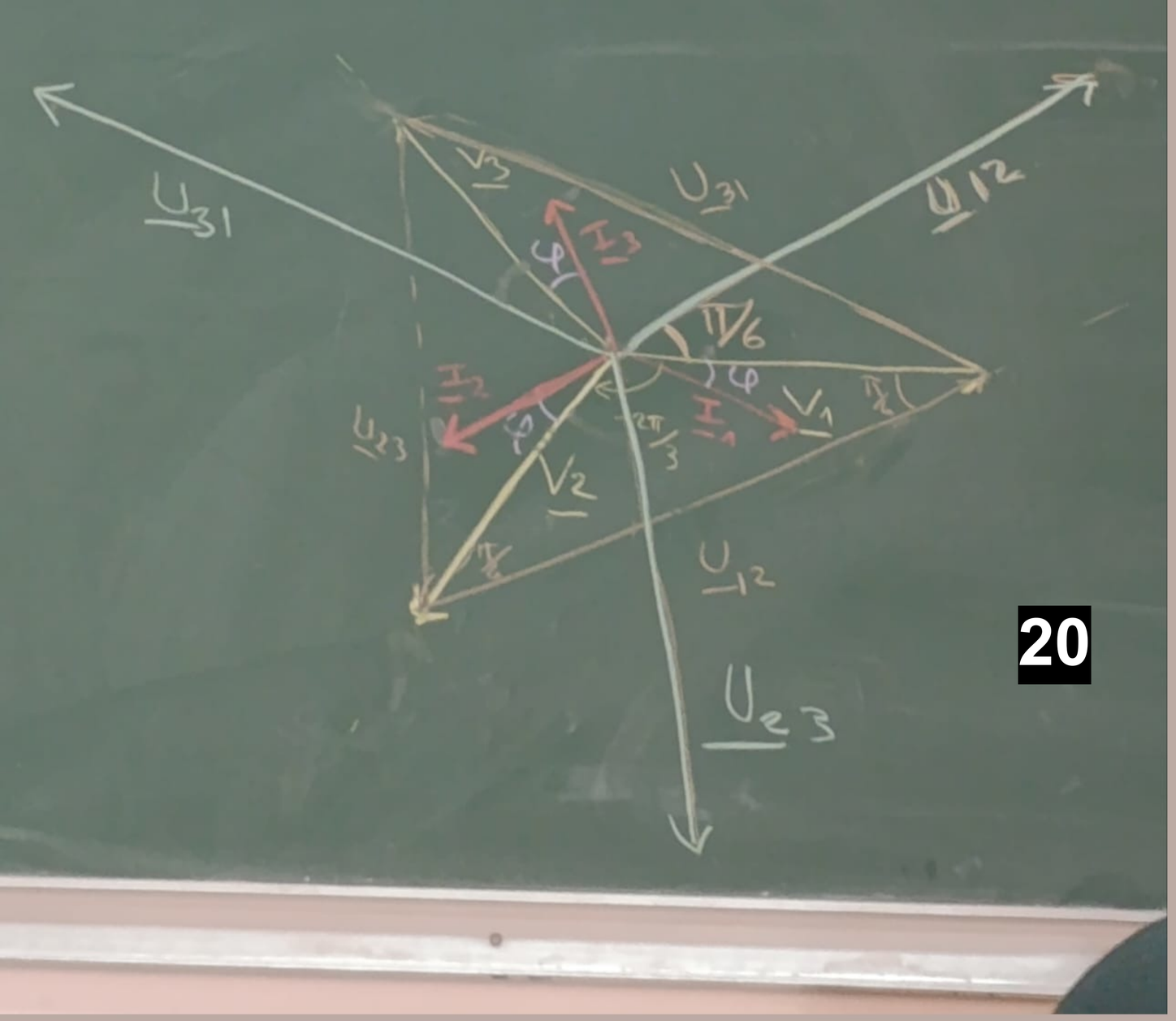
Le facteur de puissance n'est pas tolérable par le fournisseur d'énergie car il est $< 0,8$, au dessous de cette valeur le fournisseur d'energie facture

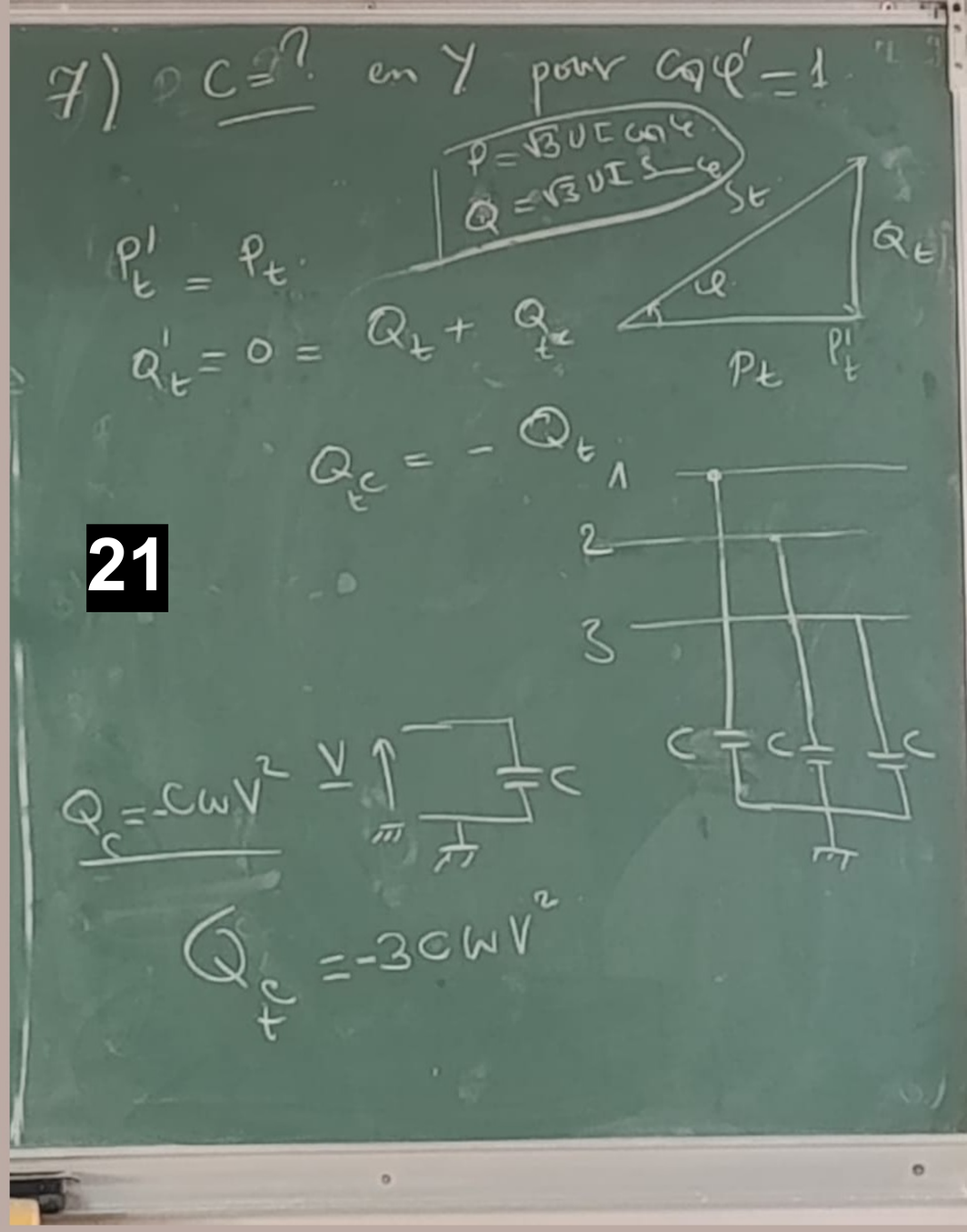
19

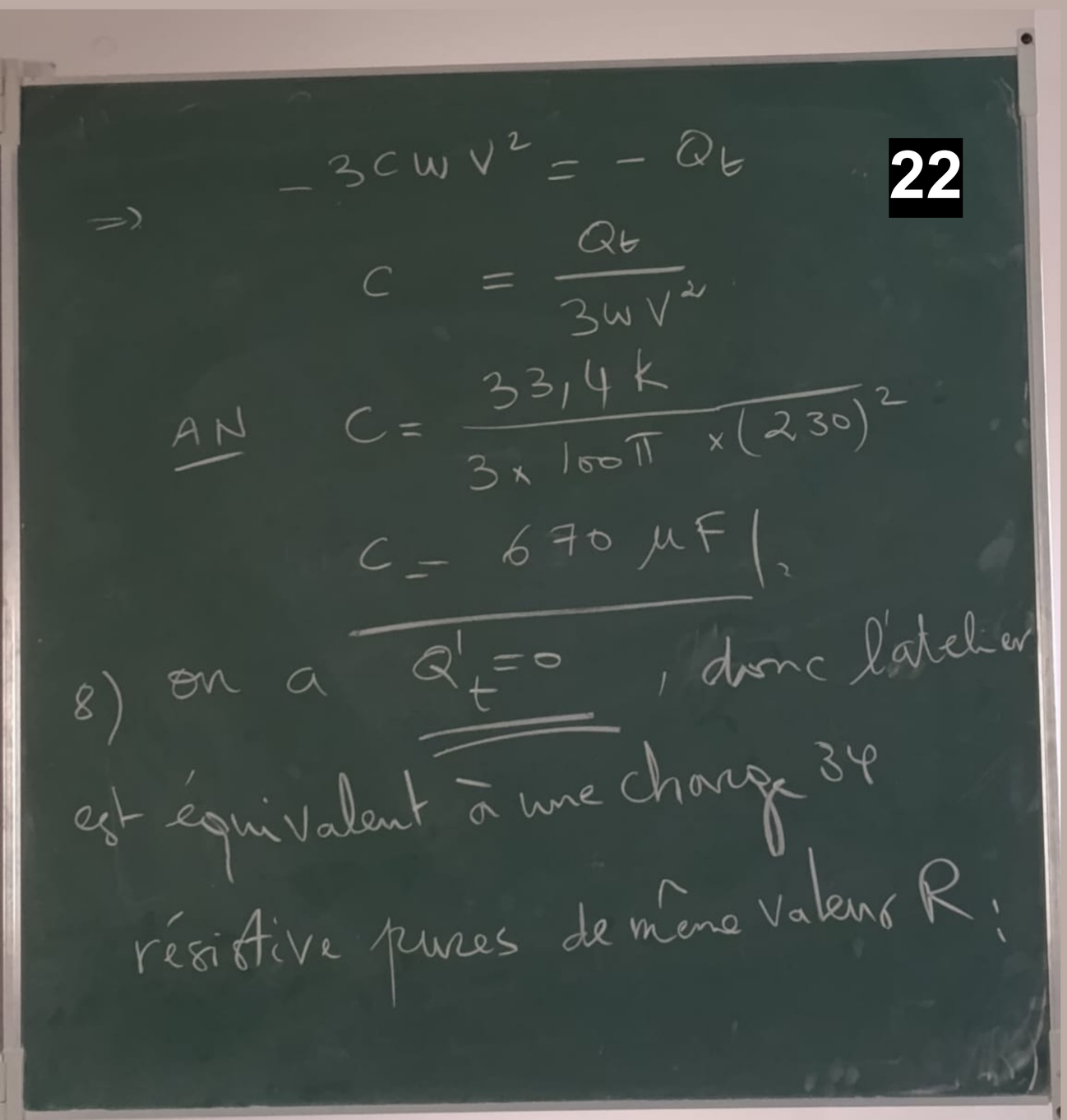
des taxes aux utilisateurs.

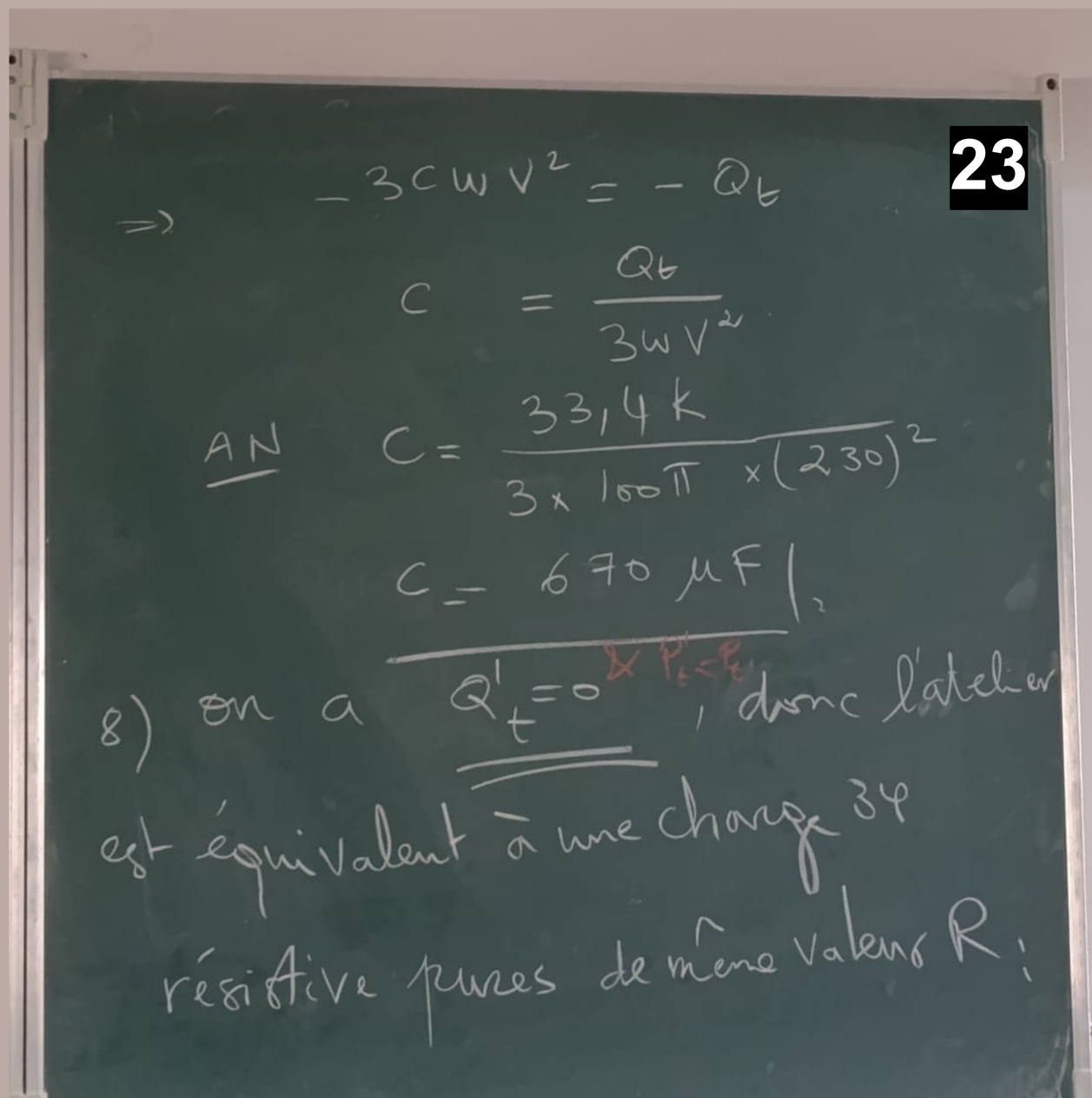
6) Représentation en plan complexes des tensions ($\underline{V}_i$, $\underline{U}_{ij}$ et les courants $\underline{I}_i$

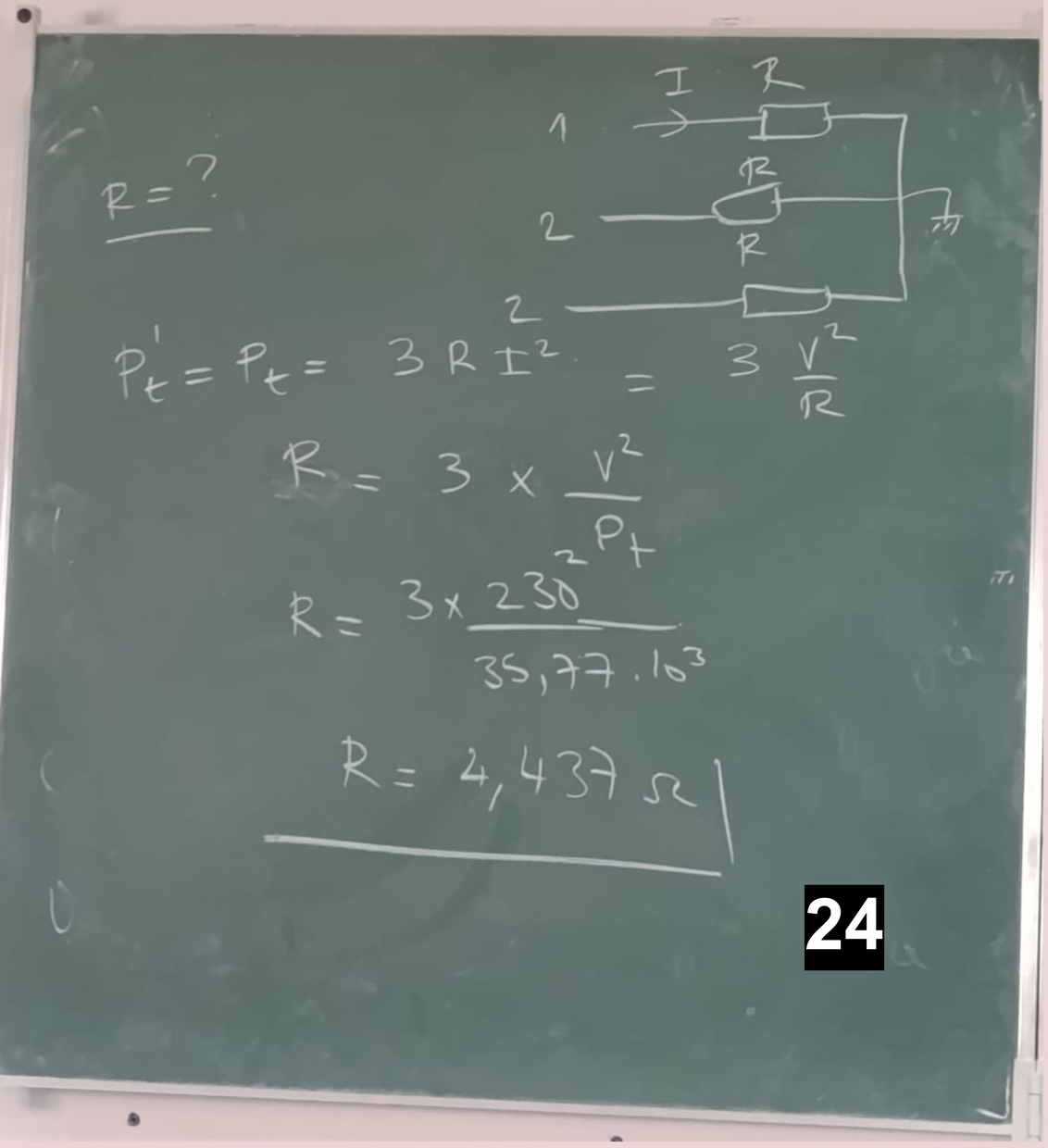



U31
V3
U31
U12
φ
I3
π/6
φ
I2
U23
φ
2π/3
I1
V1
π/2
V2
π/6
U12
U23
20


7) C=? en Y pour Cosφ'=1
P=√3 U I Cosφ
Q=√3 U I Sinφ
St
Qt
φ
Pt
P't
P't = Pt
Q't = 0 = Qt + Qc
Qtc = - Qt
21
1
2
3
C C C
Qc = -CωV² V
C
Qtc = -3CωV²


⇒ -3CωV² = - Qt
C = Qt / 3ωV²
AN C = 33,4 k / 3 × 100π × (230)²
C = 670 µF
8) on a Q't = 0 , donc l'atelier est équivalent à une charge 3φ résistive pures de même valeur R;
22


⇒ -3CωV² = - Qt
C = Qt / 3ωV²
AN C = 33,4 k / 3 × 100π × (230)²
C = 670 µF
8) on a Q't = 0 ∆ P't<Pt , donc l'atelier est équivalent à une charge 3φ résistive pures de même valeur R;
23



$R = ?$

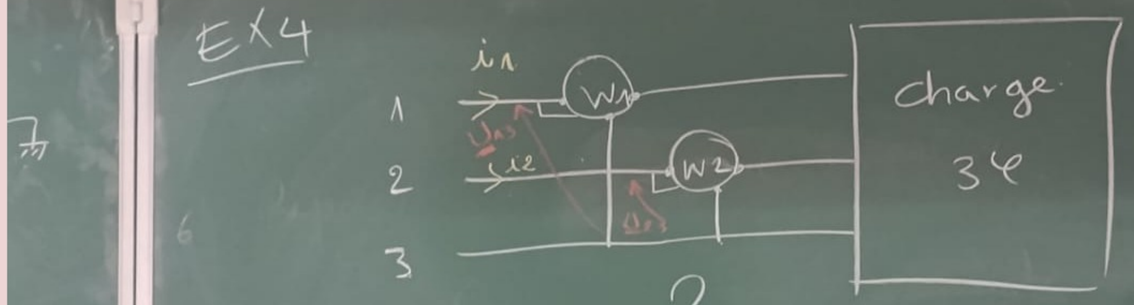
$$P_t' = P_t = 3 R I^2 = 3 \frac{V^2}{R}$$

$$R = 3 \times \frac{V^2}{P_t}$$

$$R = 3 \times \frac{230^2}{35,77 \cdot 10^3}$$

$$R = 4,437\ \Omega$$

24



$P = W_1 + W_2$?

$Q = \sqrt{3}(W_1 - W_2)$?

$W_1 = ?$, $W_2 = ?$

on a

$$\underline{U}_1 = V\sqrt{2}\sin \omega t$$

$$\underline{U}_2 = V\sqrt{2}\sin(\omega t - 2\pi/3)$$

$$\underline{U}_3 = V\sqrt{2}\sin(\omega t - 4\pi/3)$$

$$\underline{U}_{12} = U\sqrt{2}\sin(\omega t + \pi/6)$$

$$\underline{U}_{23} = U\sqrt{2}\sin(\omega t - \pi/2)$$

$$\underline{U}_{31} = U\sqrt{2}\sin(\omega t - \frac{7\pi}{6})$$

25

$$W_1 = \langle (v_1 - v_3)(t)\, i_1(t) \rangle$$

$$= \langle \underline{u}_{13}(t)\, i_1(t) \rangle$$

$$W_1 = U \times I \times \cos \theta_{U_{13}, i_1} = UI \cos(\varphi - \pi/6)$$

$$\theta_{U_{13}, i_1} = \arg(U_{13}) - \arg(i_1)$$

$$= \pi + \arg(U_{31}) - \arg(i_1)$$

$$= \pi - \frac{7\pi}{6} - (-\varphi) = -\pi/6 + \varphi$$

26

$$\underline{I}_1 = I\sqrt{2}\sin(\omega t - \varphi)$$

$$\underline{I}_2 = I\sqrt{2}\sin(\omega t - \varphi - 2\pi/3)$$

$$\underline{I}_3 = I\sqrt{2}\sin(\omega t - \varphi - 4\pi/3)$$

27

$$W_2 = \langle (\underbrace{v_2 - v_3}_{u_{23}})(t)\; i_2(t) \rangle$$

$$= \langle u_{23}(t)\; i_2(t) \rangle = UI \cos \theta_{u_{23}, i_2}$$

$$\theta_{u_{23}, i_2} = \arg(u_{23}) - \arg i_2$$

$$= -\pi/2 - (-\varphi - 2\pi/3)$$

$$= \varphi + \frac{4-3}{6}\pi$$

$$= \varphi + \pi/6$$

$$W_2 = UI \cos(\varphi + \pi/6)$$

28

29

$$W_1 + W_2 = UI \cos(\varphi - \pi/6) + UI \cos(\varphi + \pi/6)$$

$$= UI \left(\cos(\varphi - \pi/6) + \cos(\varphi + \pi/6) \right)$$

$$\cos(a-b) = \cos a \cos b + \sin a$$
$$\cos(a+b) = \cos a \cos b - \sin a$$

$$\Rightarrow$$

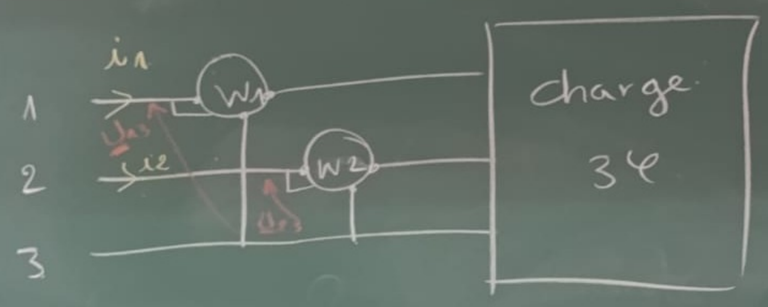
$$\Rightarrow W_1 + W_2 = 2UI \cos \pi/6 \cos \varphi$$

$\cos \pi/6 = \sqrt{3}/2$

$$= \sqrt{3}\, UI \cos \varphi$$

$$W_1 + W_2 = P$$

EX4



$W_1 =$

W_1

$\theta_{u_{13}, i_1} =$ ar

$= \pi$

$= \Gamma$

$$W_1 - W_2 = UI \cos(\varphi - \pi/6) - UI \cos(\varphi + \pi/6)$$

$$= UI \left(\cos(\varphi - \pi/6) - \cos(\varphi + \pi/6) \right)$$

$$= 2UI \sin \pi/6 \sin \varphi$$

$\sin \pi/6 = 1/2$

$$= UI \sin \varphi = \frac{Q}{\sqrt{3}} \Rightarrow Q = \sqrt{3}(W_1 - W_2)$$

30